Exercise 1: Find the *local extrema* of the function

$$f(x,y) = x^2 - 2xy + \tfrac{1}{3}y^3 - 3y \qquad (*)$$

Solution:

1) Find the critical points:

$f_x(x,y) = 2x - 2y$

$f_y(x,y) = -2x + y^2 - 3$

$f_x(x,y) = 0$ when $2x - 2y = 0 \Rightarrow 2x = 2y \Rightarrow \underline{x = y}$ (1)

$f_y(x,y) = 0$ when $-2x + y^2 - 3 = 0$

Substitute $x = y$

$-2y + y^2 - 3 = 0$

$y^2 - 2y - 3 = 0$

$(y-3)(y+1) = 0$

$y = 3$ or $y = -1$

$\Downarrow$ use (1) $x = y$ $\Downarrow$

$x = 3$ $\quad$ $x = -1$

There are two critical points:

$(3,3)$ and $(-1,-1)$

2) Use the 2nd derivative test:

$f_x = 2x - 2y$ $\qquad\qquad$ $f_y = -2x + y^2 - 3$

$\Rightarrow f_{xx} = 2 \qquad f_{xy} = -2 \qquad f_{yy} = 2y$

The discriminant is

$$D(x,y) = \begin{vmatrix} f_{xx}(x,y) & f_{xy}(x,y) \\ f_{xy}(x,y) & f_{yy}(x,y) \end{vmatrix} = \begin{vmatrix} 2 & -2 \\ -2 & 2y \end{vmatrix}$$

① $D(3,3) = \begin{vmatrix} 2 & -2 \\ -2 & 2\cdot 3 \end{vmatrix} = 2\cdot 6 - (-2)(-2) = 8 > 0$

⊕: possible max/min

$f_{xx}(3,3) = 2 > 0$

⊕

$\boxed{\text{Local min}}$

② $D(-1,-1) = \begin{vmatrix} 2 & -2 \\ -2 & 2\cdot(-1) \end{vmatrix} = 2\cdot(-2) - (-2)^2 = -8 < 0$

⊖

Saddle point

Answer: f has a saddle point at $(x,y)=(-1,-1)$
(no max/min)

f has a local minimum at $(x,y)=(3,3)$

The local minimum value is

$$f(3,3) = 3^2 - 2\cdot 3\cdot 3 + \frac{1}{3}3^3 - 3\cdot 3 = 9-18+9-9 = \underline{\underline{-9}}$$

substitute into (*)

Exercise 2: Find the *local max/min* of the function

$$f(x,y) = 2xy - x^2y - 2y^2 + 15y \qquad (1)$$

Solution:

1. Find the critical points:

$$f_x(x,y) = 2y - 2xy = 2y[1-x]$$

$$f_y(x,y) = 2x - x^2 - 4y + 15$$

$f_x(x,y) = 0$ when $2y[1-x] = 0 \Rightarrow \underline{y=0}$ or $\underline{x=1}$

$f_y(x,y) = 0$ when $2x - x^2 - 4y + 15 = 0 \quad (2)$

<u>Case 1:</u> $y = 0$. Then (2) becomes

$$2x - x^2 + 15 = 0 \Rightarrow x^2 - 2x - 15 = 0$$

$$(x-5)(x+3) = 0$$

$$x = 5, \; x = -3$$

Two critical points are $(x,y) = (5,0)$ and $(x,y) = (-3,0)$.

<u>Case 2:</u> $x = 1$. Then (2) becomes

$$2 \cdot 1 - 1^2 - 4y + 15 = 0$$

$$-4y + 16 = 0 \Rightarrow 4y = 16 \Rightarrow \underline{\underline{y = 4}}$$

A 3rd critical point is $(x,y) = (1,4)$

2. Use the 2nd derivative test:

$$[\, f_x = 2y - 2xy, \quad f_y = 2x - x^2 - 4y + 15 \,]$$

$$f_{xx} = -2y \qquad f_{xy} = 2 - 2x \qquad f_{yy} = -4$$

The discriminant is

$$D(x,y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} -2y & 2-2x \\ 2-2x & -4 \end{vmatrix}$$

① $D(\underset{x}{5}, \underset{y}{0}) = \begin{vmatrix} 0 & -8 \\ -8 & -4 \end{vmatrix} = 0 \cdot (-4) - (-8)^2 = -64 < 0$

⊖ : Saddle point

② $D(-3,0) = \begin{vmatrix} 0 & 8 \\ 8 & -4 \end{vmatrix} = 0\cdot(-4) - 8^2 = -64 < 0$

x y

⊖

③ $D(1,4) = \begin{vmatrix} -8 & 0 \\ 0 & -4 \end{vmatrix} = (-8)(-4) - 0^2 = 32 > 0$

x y

⊕

possible max/min

$f_{xx}(1,4) = -8 < 0$ ⊖: <u>local max.</u>

<u>Answer:</u> f has saddle points at $(5,0)$ and $(-3,0)$

f has a local max at $(1,4)$

The local max. value is (So, (1))

$$f(1,4) = 2\cdot 1\cdot 4 - 1^2\cdot 4 - 2\cdot 4^2 + 15\cdot 4$$
$$= 8 - 4 - 32 + 60 = \underline{\underline{32}}$$

Remark: When at a critical point, $D(x,y) > 0$, then $f_{xx}(x,y)$ and $f_{yy}(x,y)$ have the same signs. So you may use any of $f_{xx}(x,y)$ or $f_{yy}(x,y)$ to check whether there is a max or a min

[This is not true when $D(x,y) < 0$]

Assignment 1: (Due Wednesday, 9. July)

Find the local maximum and minimum values on $\mathbb{R}^2$:

1) $f(x,y) = x^3 y + 12x^2 - 8y$

2) $f(x,y) = 4xy - 2 - x^4 - y^4$

3) $f(x,y) = (1+xy)(x+y)$

Finding absolute extrema of a function on closed, bounded sets

In Calculus 1 we have learnt the Extreme Value Theorem:

If $f(x)$ is continuous on a closed interval $I=[a,b]$, the f has an absolute maximum value and an absolute minimum value on I.

Example 5: Find the absolute max/min value of

$$f(x)=x^2-2x+4 \quad \text{on} \quad [0,3].$$

closed interval

Solution: 1. Find the local max/min:

$f'(x)=2x-2$

$f'(x)=0$ when $2x-2=0 \Rightarrow 2x=2 \Rightarrow x=1$

There is one critical number:

$x=1$

Then $f(1)=1^2-2\cdot 1+4=3$

the only possible min/max value on $(0,3)$

2. Check the values of $f(x)$ at the endpoints of I:

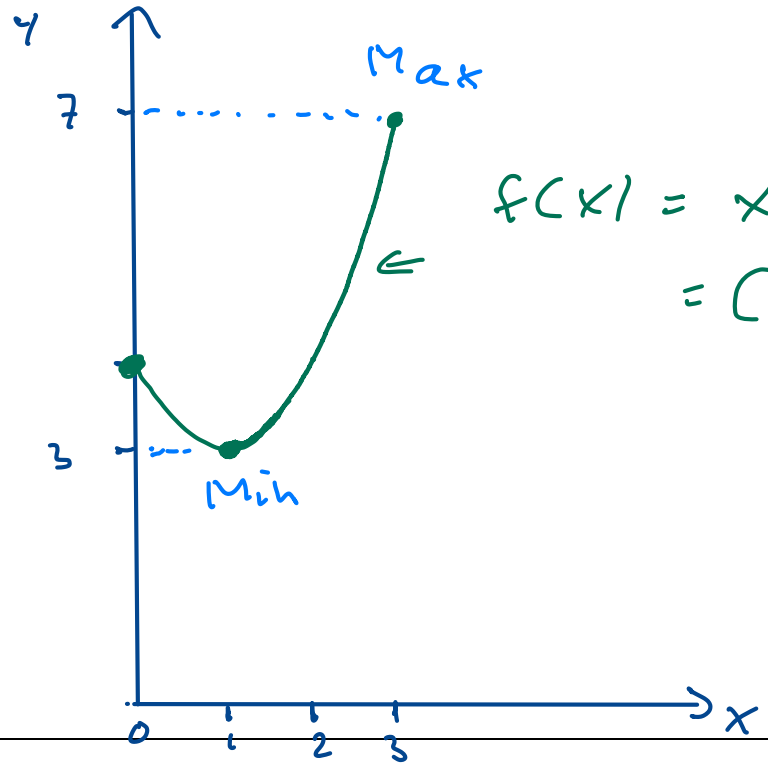
$x = 0$: $f(0) = 0^2 - 2 \cdot 0 + 4 = 4$

$x = 3$: $f(3) = 3^2 - 2 \cdot 3 + 4 = 7$

Compare the three values: Smallest: 3, largest: 7

Answer: The absolute max value is $f(3) = 7$ [at right endpoint]

The absolute min. value is $f(1) = 3$ [inside the interval $(0,3)$]



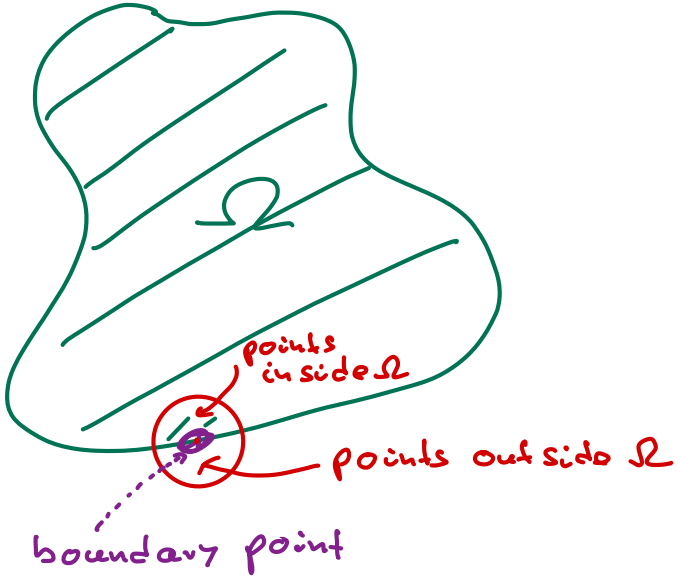
$f(x) = x^2 - 2x + 4$

$= (x-1)^2 + 3$

This generalizes to functions of 2 or more variables.

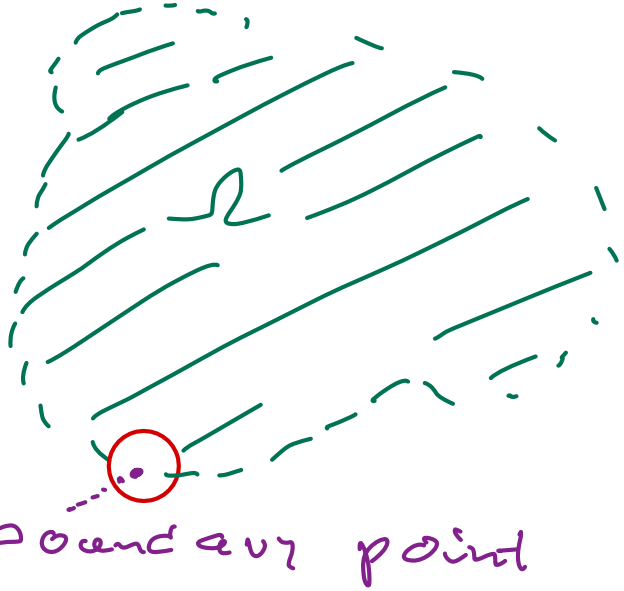
Definition: Let Ω be a subset of the plane $\mathbb{R}^2$.

1. A point (x, y) in the plane is called a **boundary point** of Ω, if <u>every disk</u> with center (x, y) contains both, points in Ω and points not in Ω.
2. The set Ω is called **closed**, if it contains all its boundary points.
3. The set of all boundary points is called the boundary of Ω.
4. The set Ω is called **bounded**, if it is contained in some (possibly very large) disk.



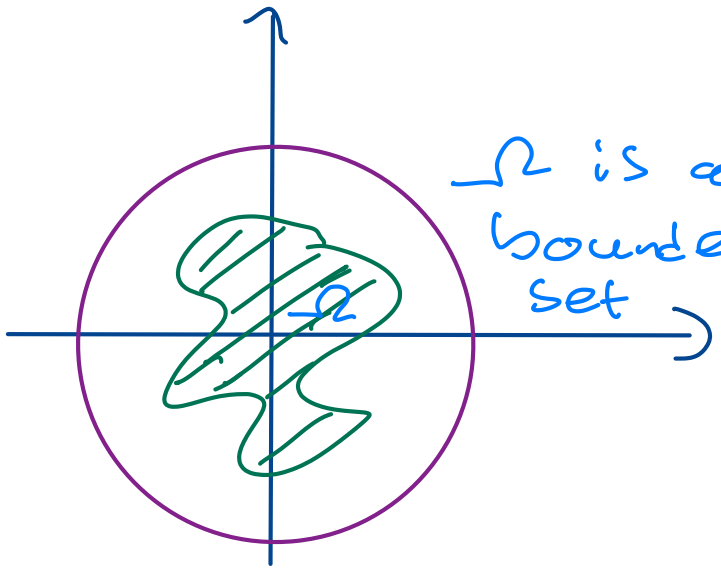
Every boundary point belongs to Ω

$\Rightarrow$ Closed Set

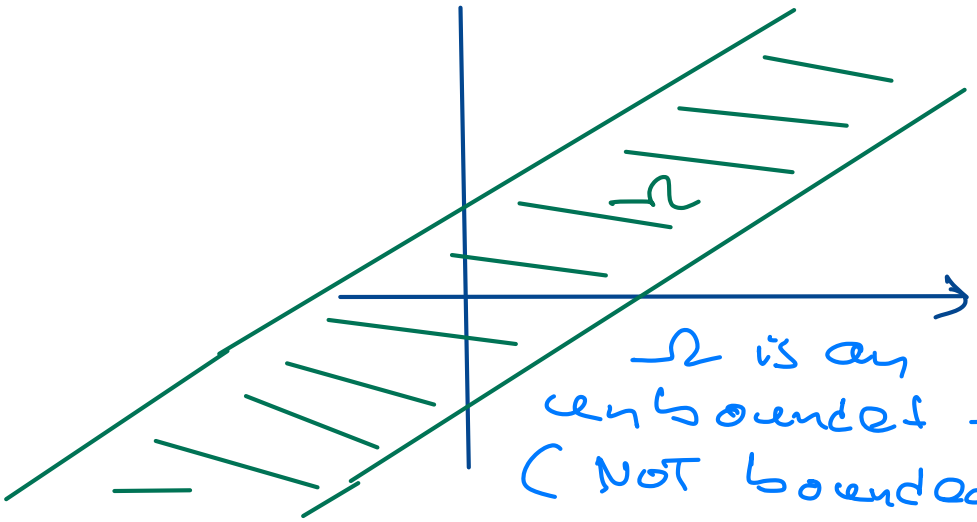


Not every boundary point belongs to Ω

$\Rightarrow$ NOT a closed set



Ω is a bounded set



Ω is an unbounded set (NOT bounded)

Theorem: (Extreme Value Theorem) If the function f is continuous on a closed and bounded subset Ω of $\mathbb{R}^2$, then f has an absolute maximum value at some point (x_1, y_1) in Ω, and an absolute minimum value at some point (x_2, y_2) in Ω.

Remark: If this point (x_1, y_1) (resp. (x_2, y_2)) is not a boundary point of Ω, then f also has a local max (resp. local min) there.

Remark: For subsets Ω of $\mathbb{R}^n$, the concept of closed and bounded set is defined similarly, and the Extreme Value Theorem still holds.

How to find the absolute maximum and minimum values of a continuous function f on a closed, bounded set Ω :

1. Find the values of f at the critical points of f in Ω.
2. Find the extreme values of f on the boundary of Ω .
3. The largest of the function values from steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

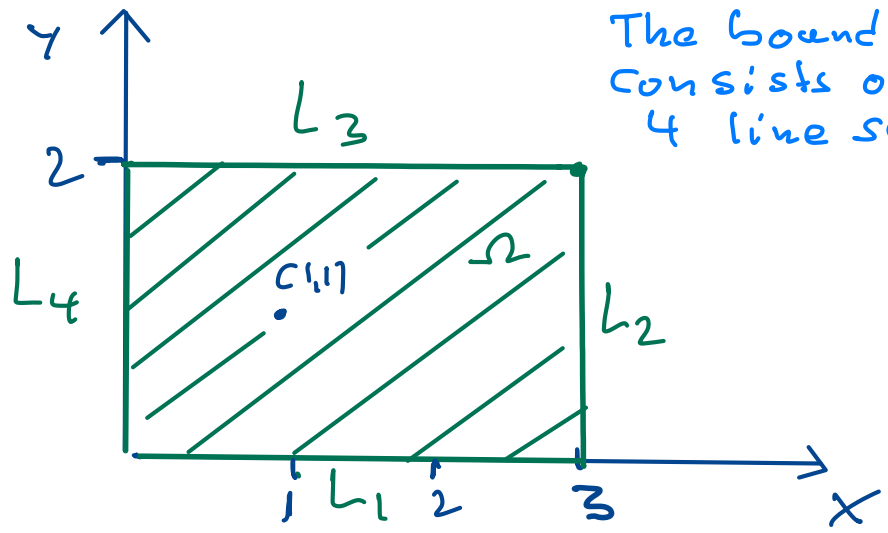
Example 5: Find the *absolute max and min values* of

$$f(x,y) = x^2 - 2xy + 2y \qquad (1)$$

on the rectangle $\Omega = \{ (x,y): \quad 0 \le x \le 3, \quad 0 \le y \le 2 \}$

Solution:

We observe that Ω is a closed and bounded set, so max/min exist.



The boundary consists of 4 line segments

1. First find the values of f

   at the *critical points inside* Ω:

$$f_x(x,y) = 2x - 2y \qquad f_y(x,y) = -2x + 2$$

$f_x(x,y) = 0$ when $2x - 2y = 0 \Rightarrow 2x = 2y \Rightarrow x = y \quad (2)$

$f_y(x,y) = 0$ when $-2x + 2 \Rightarrow 2x = 2 \Rightarrow x = 1 \Rightarrow y = x = 1$

There is one critical point: $(x,y) = (1,1)$, and $f(1,1) = 1 \qquad (*)$

2. Find max/min of f on the <u>boundary</u> of Ω:

The boundary consists of 4 line segments: L_1, L_2, L_3, L_4.

L_1 : On L_1 we have $y = 0$. Substitute into (1) and obtain

$$f(x, 0) = x^2 \quad , \qquad 0 \le x \le 3$$ (an increasing function)

Min at left endpoint
max at right endpoint

Its min. value is $f(0, 0) = 0$ // ①

Its max. value is $f(3, 0) = 9$ // ②

L_2 : On L_2 we have $x = 3$. Substitute into (1) and obtain

$$f(3, y) = 9 - 4y \quad , \qquad 0 \le y \le 2$$ (a decreasing function)

$= 3^2 - 2 \cdot 3y + 2y$

max at left endpoint
min at right endpoint

Its min. value is $f(3, 2) = 9 - 12 + 4 = 1$ ③

Its max. value is $f(3, 0) = 9 - 0 + 0 = 9$ ②

L_3 : On L_3 we have $y = 2$. Substitute into (1) and obtain

(*) $$f(x, 2) = x^2 - 4x + 4 \qquad = (x-2)^2, \qquad 0 \le x \le 3$$

(graph is a parabola)

Derivative: $2x-4$

$2x-4=0 \Rightarrow \underline{\underline{x=2}}$

Substitute $x=2$ into (*):

$f(2,2)=0$ // smallest

Its min. value is $f(2,2)=0$ (4)

Its max. value is $f(0,2)=4$ (5)

check endpoints: By (2)

$x=0 \Rightarrow f(0,2)=4$ // largest

$x=3 \Rightarrow f(3,2)=1$ //

L_4 : On L_4 we have $x=0$. Substitute into (1) and obtain

$f(0,y)=2y$, $0 \le y \le 2$ (an increasing function)

Min at left endpoint

max at right endpoint

Its min. value is $f(0,0)=0$ // (1)

Its max. value is $f(0,2)=4$ // (5)

<u>On the boundary</u>, the max. value is $f(3,0)=9$,

Compare (1)-(5)

and the min. value is $f(0,0)=0$.

3. Compare these values with (*) and see:

<u>On the rectangle</u> Ω, the absolute max. value is $f(3,0)=9$,

and inside

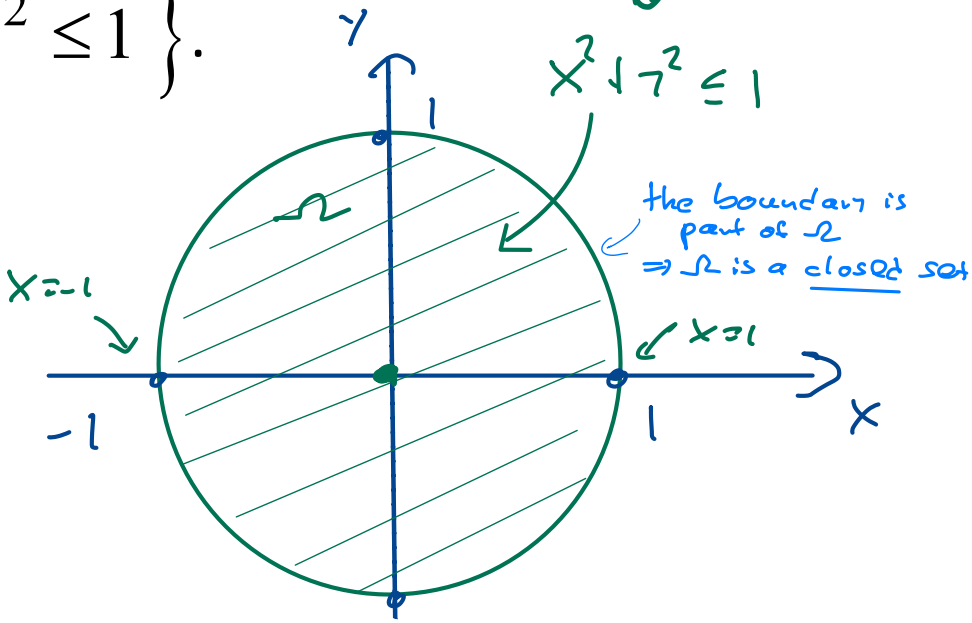
and the absolute min. value is $f(0,0)=0$. //

Example 6: Find the *absolute max and min values* of

$$f(x,y) = 2x^3 + y^4 \qquad (1)$$

on the disk $\Omega = \{ (x,y) : \quad x^2 + y^2 \le 1 \}$.

disk, radius 1
center (0,0)
$\downarrow$
$x^2 + y^2 \le 1$



Solution:

We observe that Ω is a closed and bounded set, so max/min exist.

1. First find the values of f

   at the *critical points inside* Ω:

$$f_x(x,y) = 6x^2 \qquad f_x(x,y) = 0 \text{ when } x = 0$$

$$f_y(x,y) = 4y^3 \qquad f_y(x,y) = 0 \text{ when } y = 0$$

There is one critical point inside Ω:

$$(x,y) = (0,0), \text{ and } f(0,0) = 0 \quad (*) \quad 5$$

2. Find max/min of f on the $\boxed{\text{boundary circle}}$ $x^2 + y^2 = 1$.

We get $y^2 = 1 - x^2$ Substitute into (1) and obtain on the circle

$$g(x) = f(x, 1-x^2) = 2x^3 + (1-x^2)^2 \quad , \quad \boxed{-1 \le x \le 1}$$

(This is a function of one variable !) $= 2x^3 + (1 - 2x^2 + x^4)$

We must find the max/min of $g(x) = x^4 + 2x^3 - 2x^2 + 1$ on $I = [-1, 1]$.

$$g'(x) = 4x^3 + 6x^2 - 4x = 2x[2x^2 + 3x - 2]$$

$g'(x) = 0$ when $x = 0$ or $2x^2 + 3x - 2 = 0$

$x = 0$ or $(2x - 1)(x + 2) = 0$

$x = 0$ or $x = \frac{1}{2}$ or $\boxed{x = -2}$ not in $[-1, 1]$ ✗

$g(x) = x^4 + 2x^3 - 2x^2 + 1$

Critical numbers inside $[-1, 1]$ are $0, \frac{1}{2}$

$g(0) = 1$ (1) and $g(\frac{1}{2}) = (\frac{1}{2})^4 + 2(\frac{1}{2})^3 - 2\cdot(\frac{1}{2})^2 + 1$

$= \frac{1}{16} + \frac{4}{16} - \frac{8}{16} + \frac{16}{16} = \frac{13}{16}$ (2)

Because I a closed interval, we also check at the endpoints $-1, 1$:

$g(-1) = -2$ ③ min and $g(1) = 2$ ④ max $\left.\begin{matrix} x=-1 \\ x=1 \end{matrix}\right\} \Rightarrow y = 1 - x^2 = 0$

Comparing these four values, we see: Compare ① to ④:

<u>On the boundary circle</u>, the max. value is $f(1, \overset{y=1}{0}) = g(1) = 2$

and the min. value is $f(-1, 0) = g(-1) = -2$

3. Compare these values with (*) and see: values <u>inside</u> circle $\Rightarrow$ see ⑤

<u>On the disk</u> Ω, the absolute max. value is $f(1, 0) = 2$,

and the absolute min. value is $f(-1, 0) = -2$.

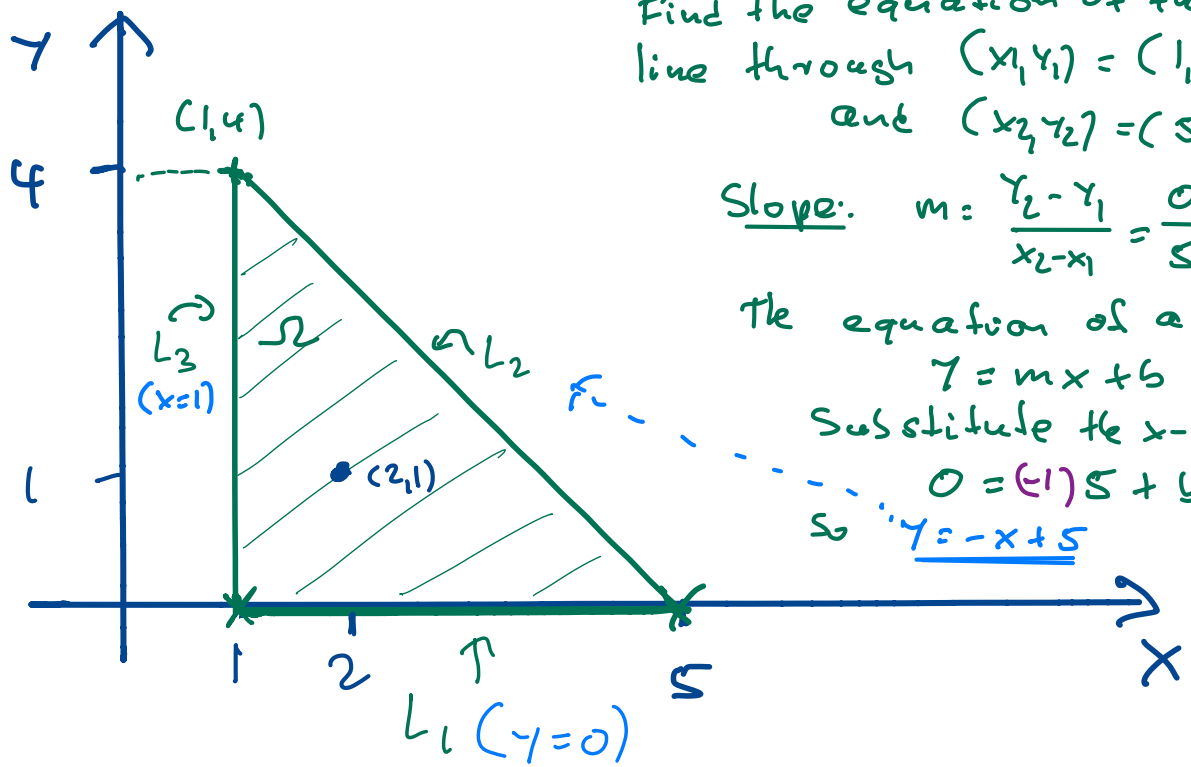
Remark: The method of Lagrange multipliers described in the next lecture can often be used to find the max/min of the function f on the <u>boundary</u> of the set Ω.

Exercise 3: Find the *absolute max/min* of the function

$$f(x,y) = 3 + xy - x - 2y$$

on the triangular region with vertices $(1,0)$, $(5,0)$ and $(1,4)$.

Solution: 1. Sketch the region.



Find the equation of the line through $(x_1,y_1) = (1,4)$ and $(x_2,y_2) = (5,0)$

Slope: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-4}{5-1} = -1$

The equation of a line is

$y = mx + b = (-1)x + b$

Substitute the x-intercept $(5,0)$

$0 = (-1)5 + b \Rightarrow b = 5$

So $y = -x + 5$ $[1 \le x \le 5]$

We see: $1 \le x \le 5$

$0 \le y \le 4$

This region Ω is a closed and bounded set.

(the boundary triangle belongs to Ω)

$\Rightarrow$ Absolute max and absolute min exist.

1. Find the critical points inside Ω.

$$f(x,y) = 3 + xy - x - 2y$$

$f_x(x,y) = y - 1 \Rightarrow f_x(x,y) = 0$ when $y - 1 = 0 \Rightarrow \underline{\underline{y = 1}}$

$f_y(x,y) = x - 2 \Rightarrow f_y(x,y) = 0$ when $x - 2 = 0 \Rightarrow \underline{\underline{x = 2}}$

There is one critical point : $(2,1)$

This point lies inside Ω $\leftarrow$ we need to check this

(see the sketch)

Then $f(2,1) = 3 + 2 \cdot 1 - 2 - 2 \cdot 1 = 1$ ①

[value of f at the critical point inside Ω]

2. Find the max/min of f on the boundary

The boundary consists of 3 line segments: L_1, L_2, L_3

$$f(x,y) = 3 + xy - x - 2y \qquad (1)$$

a) L_1: On L_1, $y = 0$. Then (1) becomes

$$g(x) = f(x,0) = 3 - x \qquad [1 \le x \le 5]$$

This is a $\searrow$ [decreasing] function

so its max is at the left endpoint of $[1,5]$

its min —— " —— right —— " ——

Max value: $g(1) = f(1,0) = 2$ (2)

Min value: $g(5) = f(5,0) = -2$ (3)

b) L_2: On L_2, $x=1$. Then (1) becomes

$$g(y) = f(1, y) = 3 + 1\cdot y - 1 - 2y = \underline{\underline{2-y}} \quad [0 \le y \le 4]$$

This is a $\searrow$ function

So its <u>max</u> is at the <u>left</u> endpoint of $[0,4]$

its <u>min</u> — " — <u>right</u> — " —

same point as in (2) above

Max value: $g(0) = f(1, 0) = 2 - 0 = 2$ (2)

Min value: $g(4) = f(1, 4) = 2 - 4 = -2$ (4)

c) L_3: The line segment L_3 has equation

$$y = 5 - x \qquad [1 \le x \le 5]$$

Substitute into (1):

$$g(x) = f(x, 5-x) = 3 + x(5-x) - x - 2(5-x)$$

$$= 3 + 5x - x^2 - x - 10 + 2x$$

(2) $$= -x^2 + 6x - 7 \qquad [1 \le x \le 5]$$

We must find the max/min of g on $[1,5]$

$$g'(x) = -2x + 6$$

$$g'(x) = 0 \text{ when } -2x + 6 = 0 \Rightarrow 2x = 6 \Rightarrow \underline{\underline{x = 3}}$$

Critical number in $[1,5]$

$y = 5 - x$

$$g(3) = f(3, 2)$$

$$\underset{(2)}{=} -3^2 + 6 \cdot 3 - 7 = 2 \quad (5)$$

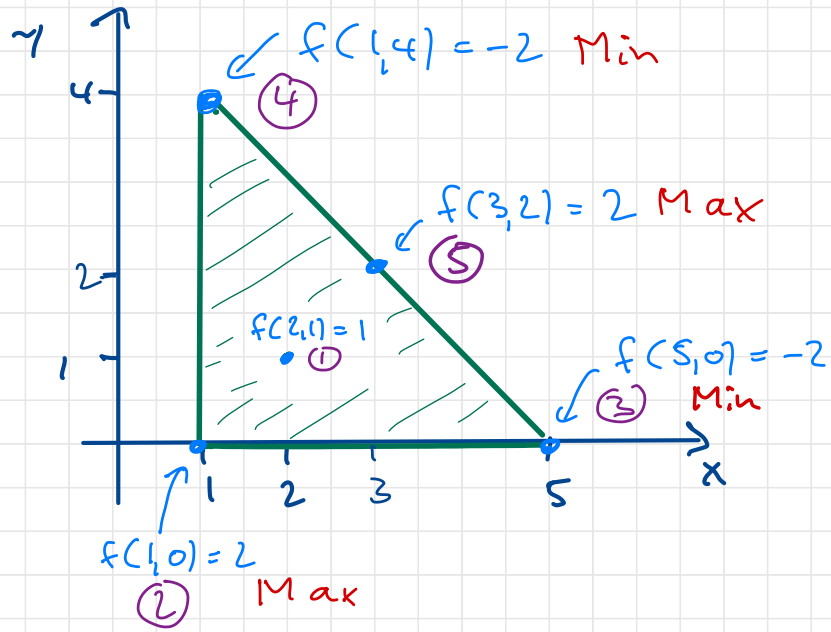
Also Consider the endpoints of $[1,5]$

However when $x = 1 \Rightarrow y = 5 - x = 4$: $(x,y) = (1,4)$

when $x = 5 \Rightarrow y = 5 - x = 0$ $(x,y) = (5,0)$

These two endpoints of L_3 are already considered in (2) and (4) above!!

let's compare all the values ① - ④



Conclusion:

a) f has its absolute max at $(x,y) = (1,0)$ and $(x,y) = (3,2)$

The absolute max value is $f(1,0) = f(3,2) = \underline{\underline{2}}$

b) f has its absolute min at $(x,y) = (5,0)$ and $(x,y) = (1,4)$

The absolute min value is $f(5,0) = f(1,4) = \underline{\underline{-2}}$